

Case Study: An Efficient Route to Evidence Safe Repurposing of Pipelines to Hydrogen

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1 ABSTRACT

Globally, networks are looking to repurpose Natural Gas Pipelines to transport Hydrogen Gas. The difference in interaction between existing network materials and hydrogen has been a key focus. Potential reductions in ductility, fracture toughness and fatigue life pose challenges with respect to pipeline design, operation, and integrity management.

In order to repurpose existing natural gas assets to hydrogen service, operators must conduct assessments to appraise their suitability and, once repurposed, implement effective integrity management programs to ensure continued safe operation. To achieve this, operators require a robust understanding of their pipeline properties, risks and susceptibility to integrity threats.

The importance of complete and accurate data to support these processes cannot be overemphasised. For many pipelines, existing integrity management systems are currently based on secondary data, often without direct reference to the supporting primary records. In order to be certain of the network capability the primary data must be identified and referenced. Hydrogen service also drives the need for new datasets to be established or expanded, with a particular focus on material performance in hydrogen environments.

This paper explores the approach taken by National Gas Transmission, supported by ROSEN, to:

- Define datasets to support appraisal and engineering assessment of assets
- Develop a digital GIS database to efficiently manage and represent data
- Rank perceived risk and completeness of data to support decision-making
- Digitise pipeline data, supported by development of automated tools to extract data from records, inspection and testing
- Optimise strategies to close data gaps using a combination of ILI, in-situ and destructive testing
- Deploy a comprehensive suite of ILI solutions to close data gaps and quantify integrity threats.

This paper will detail the progress and findings to date, with an emphasis on novel solutions and technologies developed or trialled within the project.

2 INTRODUCTION

National Gas Transmission own and operate the National Transmission System (NTS) in the UK, and are working to play a leading role in the transition to a clean energy future. National Gas Transmission, through Project Union, intends to repurpose approximately 2,000 km or 25% of the UK's gas transmission pipelines by the early 2030s, creating a hydrogen 'backbone' for the UK that connects to the European Hydrogen Backbone. National Gas Transmission are pursuing a wide array of innovation projects to advance technologies, processes and completeness of data to ensure the hydrogen transition is enacted efficiently and safely.

The NTS consists of more than 7,600 km of pipelines with diameters ranging from 100 to 1200 mm (4" to 48"). Installation dates range from the late 1960s to the present, with 45% of the total network by length installed in the 1970s, as shown in Figure 1.

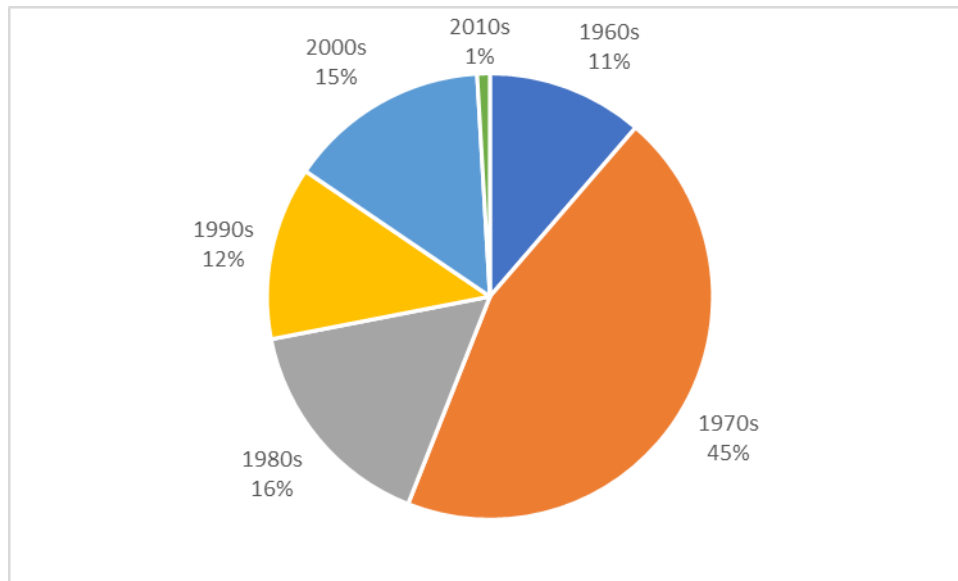


Figure 1: Breakdown of NTS Pipelines by Construction Decade

Pipeline design, manufacture and construction for the NTS were largely standardised from the early 1970s and quality assurance processes were also widely implemented. Despite the robust quality and integrity management processes that have generally been in place, there is the potential for certain threats to pose a risk to safe operation in future hydrogen service. In particular, flaws that were stable under the original hydrotest and in current natural gas service may become critical once hydrogen is introduced, due to the effects of hydrogen embrittlement.

3 EFFECTS OF HYDROGEN ON MATERIAL PROPERTIES

The presence of gaseous hydrogen is known to affect the mechanical properties of carbon steel. The principal effects of gaseous hydrogen are a decrease in ductility, a decrease in fracture toughness, and an increase in fatigue crack growth rates.

The severity of these effects reported in literature depends on several factors, principally the material properties (strength, hardness, toughness), microstructure type, hydrogen partial pressure (fugacity), test specimen types and test protocols. The effects at lower partial pressures of hydrogen may be less severe, though there is currently insufficient industry data to support quantitative statements.

- **Ductility:** The range of reduction in ductility is reported variously between 20-80% [1], [2].
- **Fracture toughness:** most sources report reductions in the range of 35-70%, although reductions up to around 85% have been reported [1], [3], [4] [5].
- **Fatigue crack growth rates** increase with hydrogen partial pressure with literature data showing increases by an order of magnitude or more [1], [6], [7].

These detrimental effects on material properties do not, in isolation, imply a risk of unsafe operation. This depends on a combination of factors, including the presence of integrity threats, the stresses acting globally on the pipeline and locally at pipeline anomalies, and the extent of reductions in material properties. To accurately assess this combination of factors and thereby the risk of failure, detailed and representative data is required.

4 DATASETS TO SUPPORT REPURPOSING ASSESSMENTS

In order to support a comprehensive repurposing assessment, detailed data is required to demonstrate compliance with codes & standards, and for inputs to engineering calculations and risk assessments to ensure safe operation with hydrogen gas. Once converted, the integrity of the pipeline must also be managed, in the context of modified threats and risks for hydrogen service, so data gathered in the repurposing stage will also be valuable in ensuring safe operation for the remaining life of the assets. The following datasets will be gathered by National Gas Transmission to support repurposing assessments and future integrity management.

- **Design characteristics:** minimum design temperature, design pressure, design factors, etc.
- **Operational data:** pressure data, gas composition data, etc.
- **Design and manufacturing:** minimum material properties, attributes and pressure ratings (for applicable components).
- **Actual material properties:** supported by manufacturing records and/or destructive testing.
- **Coating types:** external, internal, and field joint coating.
- **Hydrotest pressures:** aim pressure and actual achieved pressure by test segment.
- **Girth welds:** types and construction standards.
- **Weld procedure qualification records (WPQRs):** qualified weld properties.
- **Non-destructive examination** completed during construction or fabrication.
- **Faults, incidents and repairs:** to inform threat susceptibility and risk profile.
- **Monitoring data,** such as surveillance, direct current voltage gradient (DCVG) surveys, and close interval protection surveys (CIPS).
- **Route profile:** class locations, sensitive areas, road, rail and river crossings, including mapping of assets from records and/or inspection.
- **Inspection data:** integrity threats identified through in-line inspection (ILI), direct assessment (DA) and non-destructive examination (NDE).

Hydrogen repurposing assessments are critical undertakings. Parallels can be drawn with uprating assessments with respect to certain aspects, but there are additional complexities and risks involved that require new datasets and high confidence in the assessment inputs. Data used in such assessments should therefore be assessed in terms of quality, i.e. level of confidence and reliability of the data source.

Data relating to pipelines is often found in informal or derived sources, i.e. 'secondary' records as opposed to original 'primary' records. Where primary records are not available to support assessments, care should be taken and additional justifications may be required to support their use. Use of secondary records risks introducing inaccuracies where data has been assumed or incorrectly reproduced over time. There is also a possibility that records of construction, modifications, repairs and re-routings introduced after the original construction have been lost and different materials have been introduced with different properties and risk factors.

5 AN EFFICIENT PROCESS FOR DATA GATHERING AND ALIGNMENT

National Gas Transmission have extensive records, including original documentation from construction back to the 1960s and 70s. However, the majority of these records are archived in paper copies or microfiche films. Identifying a certain document for a specific asset from archived records is challenging, and considerable manual effort would be required to identify and extract the necessary data for hydrogen repurposing assessments.

The HyNTS Dataset project aims to create an efficient, reliable and repeatable process centred around automated and semi-automated tools that can be implemented for all assets intended for hydrogen repurposing. The end result is a complete dataset with full traceability to assets down to the level of individual pipes, components and welds, evidenced by links to supporting documentation.

An overview of the process is detailed as follows.

1. Gather data from existing databases, digital records, inspections, assessments and testing.
2. Scan paper and microfiche records.
3. Pre-processing of scanned records and apply text recognition (OCR) solutions.
4. Store documents in structured document management system with standardised metadata to support automation and search functionality.
5. Creation of accurate pipeline centreline from ILL mapping data, calibrated and checked using as-built drawings and satellite imagery. Non-pipeline assets are created from CAD files or pdf drawings.
6. Automated tools are used to extract data from records, including:
 - a. Recognition of keywords and asset identifiers
 - b. Digitised data extracted from pipeline construction records
 - c. Identification of documents with matching identifiers
 - d. Population and alignment of data to a draft 'data entry template' including automated checks for consistency of data, plausible values and formats
 - e. Hyperlinks to source documents to support the entered data.
7. Manual review of initial output provided by automated tools, including:
 - a. Enter data where automated data entry was not possible
 - b. Review and corrections of automated data entry and alignment
 - c. Finalise data classification as primary or secondary.
8. The final 'data entry template' is integrated to the UPDM geodatabase structure in ArcGIS by use of automated scripts.
9. Alignment and integration of data to the geodatabase from inspection and testing, via linear referencing and/or spatial co-ordinates.

The main aspects of the process are illustrated in Figure 2.

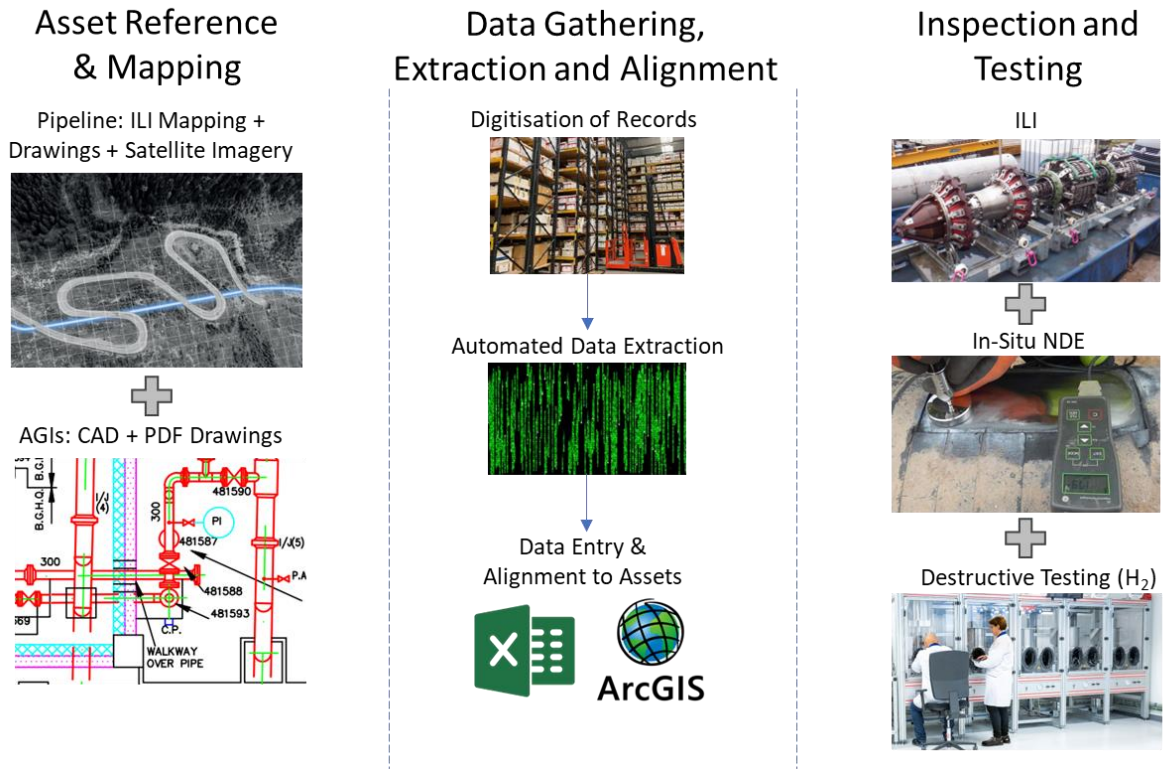


Figure 2: Illustration of Asset Mapping, Data Gathering and Geodatabase Creation, including Alignment of Inspection & Testing Data

6 THE VALUE OF GIS SOLUTIONS

Geographic Information Systems (GIS) provide a solution that combines a detailed database model with spatial mapping of assets. Pipelines and other assets can be represented down to the level of individual pipes, welds, components, and so on, and detailed data can be aligned against each of these features.

Solutions based on UPDM or PODS provide base schemas with data fields for many aspects of pipeline data that can be used to support integrity management and safe operation. However, many types of detailed data are not found in the base schemas, for example relating to toughness properties, and fields specifically relating to hydrogen service are not represented in current schemas. Extensions to these schemas were therefore defined to enable the full extent of data discussed in Section 4 to be captured. Additionally, layers were added to enable classification of each data entry (i.e. primary or secondary) and referencing of source data via direct hyperlinks to evidence the accuracy of data.

Development of a detailed and extended GIS database enables multiple types of data to be represented against a single centreline with accurate spatial and linear referencing. Information relating to the pipeline route, material properties and integrity threats can be viewed in parallel and queried for particular values, status and completeness of data with full granularity. Use of dashboard views, such as illustrated in Figure 3, can provide useful visual summaries of the data within a selected area. In this example, the pipeline centreline is shown in yellow, with open circles showing girth welds, and coloured circles showing the locations of metal loss and geometrical anomalies.

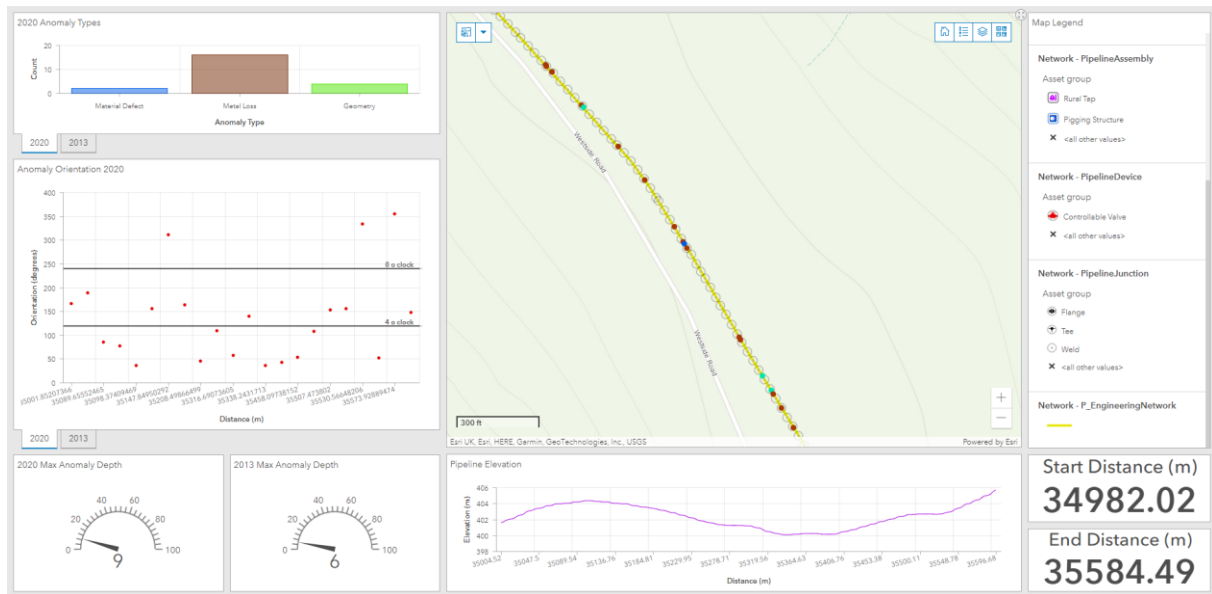


Figure 3: Example Dashboard View Displaying GIS Data and Metrics

7 AUTOMATION AND EFFICIENT PROCESSES FOR DATA ENTRY

A pipeline segment with a length of 50 km, for example, will contain several thousand individual pipes, components and welds. It is common for construction of pipelines and associated assets to use pipe and components from multiple manufacturers. Properties and attributes specified in design may also differ for example due to class location changes, crossings and building proximity. This results in a significant task to define the different materials and weld types present within a given pipeline segment, assign specific properties and attributes, and identify the relevant manufacturing or construction records. Within a robust process, traceability should be evidenced between the components and the original manufacturing and construction records.

An example of construction records for a National Gas pipeline is shown in Figure 4. Weld identifiers are listed along with records of NDE completed for each weld. Properties and attributes are also recorded for each individual pipe and component. Pipe/component numbers are listed with traceability to specific heat (cast) numbers, which enable manufacturing records (MTRs) to be linked to specific pipes. The example in Figure 4 represents a small extract from data that can extend to thousands of welds, pipes and components; it would be a significant task to manually align this data to the assets and populate a database.

For high quality records with typed data, such as the example in Figure 4, automation is achievable to recognise, digitise and tabulate the heat (cast) numbers corresponding to each individual pipe with high success rates. Handwritten documents and scanned copies with inconsistent quality and formatting pose greater challenges, requiring additional steps in pre-processing and more advanced optical character recognition (OCR) solutions; such processes will require a greater proportion of manual effort and review of the automated content extraction.

JOINT REJECTED (REPAIR ETC)								
JOINT PHASE ARRAY								
JOINT ULTRASONICS								
JOINT MPI			08-08-16				08-08-16	
WELD No			GW1	W31	W27	W26	GW2	W
PIPE No			25A	37	1	95	10	
CAST No			14-01-76	429542	1NU342	429542	429542	
LENGTH (M)			2100	4340	n/a	9500	12422	
WALL THICKNESS (MM)		RDX side	19.1X12.7	19.1	19.1	19.1	19.1	
GRADE		Existing Pipe	L415 ME	L415 ME	L415ME	L415 ME	L415 ME	
DESCRIPTION (PIPE/BEND°)		Feeder 6	762mm Pipe	762mm Pipe	45° Bend	762mm Pipe	762mm Pipe	

Figure 4 – Example of Construction Records Detailing Completed Weld NDE, Material Heat (Cast) Number, Wall Thickness and Grade for Each Weld and Individual Component

The heat (cast) number 429542 identified from the construction record in Figure 4 is found to relate to an available MTR, as shown in Figure 5, via OCR and search functions. The MTR provides the yield strength, ultimate tensile strength, elongation, hardness, mill test pressure and Charpy data for the pipe body, weld and heat-affected zone (HAZ). These data are valuable in evidencing compliance with material restrictions for hydrogen service, such as those specified in IGE/TD/1 Supplement 2 or ASME B31.12. Additionally, the Charpy values may be correlated to fracture toughness values and used to justify conservative assumptions for in-hydrogen performance (as permitted by IGE/TD/1 Supplement 2). If an anomaly were to be identified in these pipes, for example by ILI or direct assessment, then these data can now be used directly in the engineering assessment with high confidence.

B=Body W=Weld L=Longitudinal T=Transverse S=Stripped Out Section C=Round Bar KV=Charpy Impact CL=Centre Line FL=Fusion Line F2=F/Line+2mm F5=F/L M=Mid-Thickness R=Root E=Absorbed Energy F=Fibrosity HV=Hardness VPN HR=Hardness Rockwell DT=Drop Weight Tear Test A=Annealed N=Normalised NT=No																			
B07 Cast/Heat No.	Pipe No.	CO0 Test Piece No.	Tensile Test										Impact/Hardness Tests						
			C01	C02	C10	C11 Yield Stress Re N/MM2	C12 Tensile Stress Rm N/MM2	C13 Elong A% S.65	Ratio Re/ Rm	B05	D02 Hydro Pressure BAR	C01	C02	C40 C30 Test Type	C41 Width (mm)	C03 Temp (°C)	Values J/%		
429542	DLN 00106		B	T	S	424	593	24	72		187	BO	T	KVE	10.0	0	203	233	232
			W	T	S	496	613							KVF			100	100	100
												CLO	T	KVE	10.0	0	183	191	192
														KVF			100	100	100
												HAZ	T	KVE	10.0	0	210	206	196
														KVF			100	100	100
												B	T	DT		0	100	100	100

Figure 5 - Example Material Test Record (MTR) Corresponding to Pipes in Figure 4

This set of data has been used to demonstrate the approach at this initial stage of the project. OCR followed by implementation of Python scripts enabled the 'bar chart' shown in Figure 4 to be digitally tabulated, giving an extracted listing of weld numbers, associated component types, material grade, and cast numbers. Automated approaches can then identify matching cast numbers within a set of pdfs from original manufacturing records, such as the MTR shown in Figure 5, and match these against the applicable pipes and components. A hyperlink to the file is then also populated within the geodatabase, alongside the applicable component, to evidence the data entered.

This approach significantly reduces manual effort in typing data from records to create a tabulated listing. Automatic recognition of the manufacturing records associated with each cast number then saves further manual effort, which would otherwise consist of reading through and cross-referencing hundreds of files.

Note that this example is given as a limited demonstration, at this initial stage of the project, with further automation intended to incorporate other sources of data. The level of automation implemented will be balanced, in terms of development effort contrasted with reasonable manual tasks depending on the document types and complexities in data extraction.

8 INSPECTION AND TESTING

8.1 A COMBINED APPROACH

Despite thorough records searches, it is inevitable that there will be data gaps. Hydrogen service also drives the need for new datasets to be established or expanded. In particular, there is a focus on material properties in hydrogen gas environments, as well as increased risk posed by the possible presence of cracks and crack-like defects. Hard spots are also identified as a possible threat with increased relevance in hydrogen, since their high hardnesses and microstructure types mean they may provide initiation points for cracking [8], [9].

In order to detect and quantify integrity threats in the context of future hydrogen service, National Gas Transmission have defined a comprehensive approach incorporating advanced ILI solutions, in-situ NDE and destructive testing as summarised in Table 1. For buried pipelines, ILI is the most efficient way to achieve extensive data coverage, supplemented by validation primarily using in-situ NDE. For above-ground assets that are not piggable, NDE can be implemented directly. Cutouts for destructive testing can also be used where required to further characterise cracks and crack-like anomalies, as well as providing material for destructive testing in air and in hydrogen gas environments.

Table 1 – Inspection and Testing Approach for Hydrogen Repurposing

Type	Data	ILI Technology	Validation / Non-Destructive Testing (In-Situ)	Destructive Testing
Anomalies	Metal loss - volumetric	MFL-A	Visual examination / laser scan	-
	Axial cracks	EMAT-C + MFL-C	MPI + UT RT / CT	Sectioning
	Geometric anomalies	XT (Caliper)	Visual examination + laser scan	-
Pipeline Route	Mapping	XYZ (IMU)	-	-
	Depth of cover	XYZ (IMU)	Large stand-off magnetometry (LSM)	-
Materials	Pipe grade	RoMat PGS	Chemical composition + hardness (estimate) or IIT	Tensile testing
	Pipe populations (material grouping)	RoMat PGS	Chemical composition + hardness	Tensile testing + chemical composition + microstructure
	Pipe / weld type	MFL-A / MFL-C	Visual examination	-
	Hard spots	RoMat DMG	Handheld magnetometer Portable hardness testing	Microstructure Hardness testing
	Toughness	-	-	Testing in H ₂

8.2 FINDINGS FROM ADVANCED ILI PILOT PROJECT

A comprehensive inspection campaign using the full suite of ILI technologies listed in Table 1 has recently been completed for a 36 km long, 30" (750 mm) diameter pipeline segment within the NTS, with inspection of further pipeline segments intended over the coming years as aligned with the overall hydrogen strategy. Discussion of findings from the 'advanced' ILI services is given below.

EMAT-C: only two anomalies were identified from the inspection, classified as 'linear anomalies' (potentially crack-like). Validation will be completed to enable final classification and sizing of these

anomalies. Aside from these two reported anomalies, there were no reported features in the pipeline segment. Once validation (and remediation, if required) of the reported anomalies is completed, the absence of other detectable anomalies significantly de-risks the presence of crack-like anomalies that would pose a threat to safe operation in hydrogen service. Completion of the EMAT-C inspection enables use of the service specification to support assumption of the maximum existing crack sizes within the pipeline for quantitative ECA calculations.

RoMat DMG: hard spots were reported associated with 10 separate pipes, with a maximum reported hardness of 264 HV. This is not considered a high enough hardness to be a direct integrity threat. However, validation may be considered on an opportunistic basis to validate the ILI findings and de-risk the potential for local high hardness regions that could present potential sites for initiation of cracks in hydrogen service.

RoMat PGS: A total of five pipe ‘populations’ (materials) have been reported within the inspected pipeline segment. The identified populations evidence that the majority of the pipeline consists of the original 1972 construction using two different wall thicknesses (11.91 and 12.70 mm) termed populations A1 and B1. The consistent RoMat PGS sensor values within these populations support the conclusion that material properties and metallurgical characteristics are consistent and therefore expected to demonstrate comparable behaviour in hydrogen gas environments.

Two individual pipes were identified as belonging to different populations, as identified by changes in reported strength and sensor response. Review of satellite imagery and records identified these as pipes installed to remove block valves in 2011; material properties and metallurgical characteristics are fully documented for these two pipes.

Additionally, an outlier was identified with slightly lower reported strength within the ‘B1’ population. This may represent differences in chemical composition or metallurgy compared with the surrounding pipe materials, and therefore may demonstrate different performance in hydrogen. If material is removed for destructive testing to establish material properties in hydrogen gas, it may be recommended to target sampling at this location to ensure the full range of material properties are quantified.

8.3 VALIDATION AND FURTHER TESTING

The combined ILI results support development of a clear and optimised strategy to complete data gathering and support the repurposing assessment. Definition of populations through the RoMat PGS service provides a clear breakdown of the pipeline into equivalent material types, ensuring all materials are characterised and providing precise sampling locations for further testing when anomalies are aligned with the populations, as shown in Table 2. In particular, in cases where cracks or crack-like anomalies are identified within specific populations, toughness data can be specifically gathered from the affected population(s) to support ECA calculations.

Table 2 – Anomalies from ILI Aligned with Pipe Populations

Population	Total Length (m)	% of Total Length	No. Pipes	Corrosion Max. Depth (%)	Cracks / crack-like Anomalies	Hard Spot Indications
A1	24,661	68.5%	2,181	32%	-	8
A2	14	< 0.1%	2	-	-	-
B1	11,293	31.4%	1,074	25%	2	2
B2	10	< 0.1%	1	-	-	-
C1	42	0.1 %	14	15%	-	-

For illustration, in this case five sampling locations would permit comprehensive validation addressing the two linear (potentially crack-like) anomalies, three hard spot anomalies with different observed characteristics, and provide access to material from all four populations in the pipeline (excepting population C1 in the launcher/receiver as above ground). The identified populations are overlaid with the reported anomalies from ILI, and illustrative sampling locations, in Figure 6.

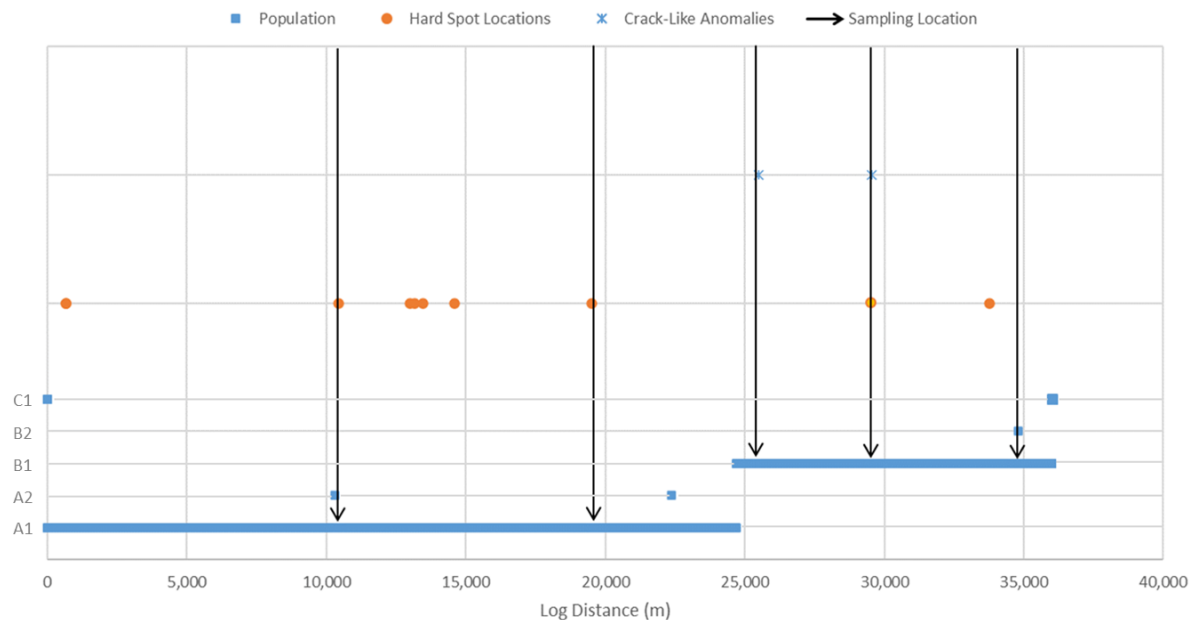


Figure 6 – Pipe Populations with ILI Anomalies Overlaid and Illustrative Sampling Locations for Validation or Further Testing

8.4 BRINGING IT ALL TOGETHER

As described in Section 5, the ILI data gathered from the completed inspections detailed in this section will be imported to the UPDM geodatabase as additional data layers. Once data gathering and extraction has been completed from records for this pipeline section, these datasets will be aligned with the same centreline and asset referencing. Any further data gathered from inspection and testing will also be aligned with the same reference on an ongoing basis.

These datasets can then be used in combination to assess the state of data completeness and readiness to support a formal hydrogen repurposing assessment. The repurposing assessment will benefit from complete data on all relevant integrity threats and the underlying pipeline data necessary to support robust and evidence-based compliance, engineering and risk-based assessments. The data will also facilitate ongoing integrity management with detailed and high quality data for the remaining life of the pipeline.

9 CONCLUSIONS

Hydrogen repurposing assessments require new datasets to be established, with particular importance placed on material properties in hydrogen environments and increased focus on particular integrity threats, especially the presence of crack-like anomalies and hard spots.

To enable robust and representative engineering and risk-based approaches, data used in repurposing assessments should be of high quality and evidenced by records, inspection and testing. National Gas Transmission have extensive records, including original documentation from design, manufacture and construction spanning more than 50 years; however, the majority of these records are archived in physical formats and identifying specific source data is challenging and time consuming.

The work described in this paper will implement robust processes and tools to gather, digitise, align and map data to assets. In order to meet the ambitious timelines set by National Gas Transmission for hydrogen repurposing, this process must be both efficient and scalable across an extensive asset base, without sacrificing accuracy or rigour. To achieve this, automated tools are under development to digitise and extract data from scanned records and existing databases, and subsequently populate a geodatabase with full granularity down to each pipe and component (with manual engineering review prior to finalising datasets).

In parallel with these data gathering efforts, a comprehensive scope of inspection and testing is underway to detect and quantify integrity threats relevant to future hydrogen service, and to measure material properties in hydrogen gas environments. This approach centres around advanced ILI including crack detection and material property services, supplemented by NDE and destructive testing. Completion of this advanced ILI scope on a pilot pipeline segment has demonstrated its value in identifying, and ultimately mitigating and de-risking, the presence of potential integrity threats that would otherwise have been unknown factors. The combined datasets provide robust quantitative inputs to repurposing assessments and support efficient strategies for further inspection and testing.

By 2025, this strategy for data gathering, testing and inspection will be completed for several hundred kilometres of pipelines and dozens of stations, providing high quality, evidence-based data to facilitate repurposing assessments and meet the ambitious timelines set by National Gas Transmission for introduction of hydrogen to the UK National Transmission System.

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